KUNJAL MAHAJAN

AI & Data Science Engineer · Computer Vision, Deep Learning & GenAI

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SUMMARY

Final-year B.E. student in Artificial Intelligence & Data Science who builds applied computer vision, deep learning, and NLP systems. Built BlinkBridge AI, a real-time system that turns eye blinks and head gestures into Morse code, text, and speech, and a CNN-based tuberculosis detection system for chest X-rays. Currently learning full-stack development with AI through Finlatics.

EXPERIENCE

Founder · BlinkBridge – Blink to Morse *Ongoing*

- Founding and building BlinkBridge, an assistive communication tool that lets people who cannot speak or type communicate using only eye blinks and head movements, captured through a standard webcam.
- Designed a modular Python architecture with separate components for blink detection, head-gesture recognition, Morse translation, calibration, speech output, and session logging, so each part can be tested and improved independently.
- Built an auto-calibration step so detection adapts to each user's eyes and camera setup, added a countdown-based interface that guides each input, and integrated text-to-speech so decoded messages are spoken aloud.

PROJECTS

BlinkBridge AI – *Bridging silent communication through intelligent eye blinks* *Tools: Python, MediaPipe, OpenCV, pyttsx3*

- Real-time computer vision system that detects eye blinks, head nods, and head turns and converts them into Morse code, which is decoded into text.
- Used MediaPipe face tracking and OpenCV to detect blinks in real time, and head pose estimation with cv2.solvePnP to recognise nods and turns as additional input gestures.
- Added a Morse-to-text translator, pyttsx3 text-to-speech output, session logging, and a live countdown UI.

AI-Based Tuberculosis Detection from Chest X-Ray Images *Tools: Python, Deep Learning, CNN, Medical Image Processing*

- Developed a deep learning system that detects tuberculosis from chest X-ray images using CNN-based medical image analysis, supporting early detection.

AI-Based Speech-to-Text and Language Translation Engine *Tools: Python, Speech Recognition, NLP, Translation APIs*

- Built an AI system that converts English speech to text and translates it into Hindi, Marathi, Korean, Japanese, Turkish, and Spanish using speech recognition and NLP techniques.

Inventory Management & Billing System *Tools: Python, MySQL*

- Built a Python and MySQL system that automates inventory tracking, billing, and report generation, with a structured, user-friendly interface for stock and sales management.

Hangman Game Using Python *Tools: Python*

- Built an interactive Hangman game in Python with clean game-flow logic and a user-friendly interface.

TECHNICAL SKILLS

GenAI & LLM: RAG pipelines, prompt engineering, OpenAI, Gemini, Claude, ChatGPT
Machine Learning & Engineering: Deep learning, ML statistics, Feature Engineering, Model Evaluation, PyTorch, statistical hypothesis testing, NumPy, Pandas, TensorFlow, Python, TypeScript/Node.js, SQL, GitHub Actions, Weights & Biases, Git
Backend & Infrastructure: Python (NumPy, Pandas), TypeScript/Node.js, SQL, GitHub Actions
Programming Languages: C, C++, Python, Java, SQL
Web Technologies: HTML, CSS
Database Management: MySQL (DBMS)
Data & Analytics Tools: MS Excel
Libraries / APIs: Python Libraries, Speech Recognition APIs, NLP & Translation APIs
Operating Systems: Windows

EDUCATION

B.E., Artificial Intelligence & Data Science · *Terna Engineering College, Navi Mumbai (University of Mumbai)* *2023 – 2027 (Pursuing)*

HSC (Science) · *Heritage Junior College, Nanded (MSBSHSE) · 63.17%* *2023*

SSC · *Kids' Kingdom Public School, Nanded (CBSE) · 84.6%* *2021*

CERTIFICATIONS & HACKATHONS

Full Stack Development with AI, Finlatics (In progress) | Google Data Analytics Course, Coursera | Build and Grow AI Hackathon 2.0, GDG Cloud Mumbai, 2026

ACHIEVEMENTS

Miss Fresher 2023, Terna Engineering College | National Level Championship Winner, Kathak, Folk and Contemporary Dance | State-Level Classical Dance Artist | Yogist, State Champion | Head of College Dance Crew, Terna Engineering College | Ex-Publicity Head, E-Cell | Ex-Co-Secretary, Department Magazine Committee

EXTRA-CURRICULAR ACTIVITIES

Dancing, singing, painting, sketching, writing, yoga, karate, modeling, photography, fashion designing, sports, archery & shooting, cooking.